



wazuh.

WAZUH FIM + ACTIVE RESPONSE

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[Portfolio](#)

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1. Introduction

In the previous tutorial, we used [File Integrity Monitoring \(FIM\) with VirusTotal](#) to identify whether a newly created or modified file is safe or malicious. While this feature provides strong visibility, detection alone is not enough but malicious files can still be executed or spread before an analyst reacts.

To enhance protection, Wazuh allows us to use **Active Response**. With this feature, the system can automatically act when VirusTotal reports that a file is harmful. In this tutorial, we configure Wazuh to automatically **delete any malicious file** detected by FIM and confirmed by VirusTotal. This turns Wazuh from a monitoring tool into an automated threat-response system, improving security and response.

2. Benefits of Using Active Response with VirusTotal

Using Active Response along with VirusTotal adds an extra layer of automated protection to the environment. The key benefits include:

- Malicious files detected by VirusTotal are immediately deleted without waiting for manual investigation.
- The system reacts within seconds, reducing the risk of malware execution or lateral movement.
- Routine malware cleanup is handled automatically, so that the analysts may focus on more important tasks.
- Active Response ensures that known malicious files are eliminated before they can cause any impact.

3. How Active Response Works in Wazuh

The workflow for this active response is straightforward:

- **FIM detects** a new/modified file.
- Wazuh sends the file hash to **VirusTotal**.
- If VirusTotal finds malware → **Rule ID 87105** triggers.
- Active Response listens for this rule.

- When triggered, it executes your script (delete or quarantine the file).

Learn more about response automation here: [Wazuh Active Response](#)

How Active Response Works in Wazuh

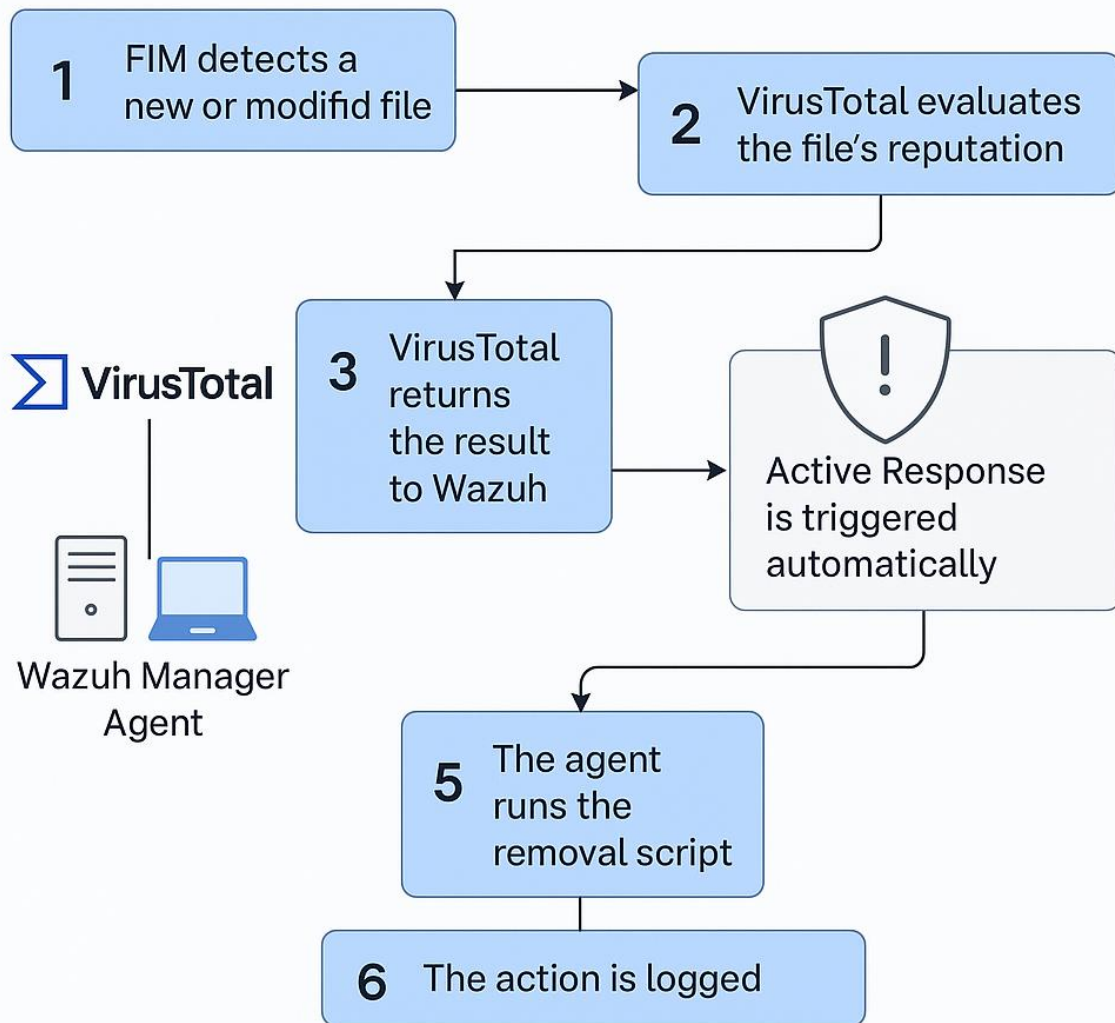


Figure 1: Active Response workflow

This automation ensures a clean, efficient workflow that enhances security.

4. How to Configure Automatic Malware Deletion

In this section, I configured Wazuh to automatically delete any file that VirusTotal identifies as malicious. The setup requires three main components:

1. **FIM** monitoring of a directory
2. **VirusTotal** integration on the Wazuh Manager

3. **An Active Response script** that removes the malicious file (remove-threat.sh)

In the next paragraphs, I will walk through to configure both **Wazuh Manager** and the **Wazuh Agent** to enable this automation.

4.1 Enable FIM Monitoring on the Agent

On the Wazuh agent, edit the agent configuration file (/var/ossec/etc/ossec.conf) inside the <syscheck> block if you want FIM to monitor downloaded files.

```
<directories realtime="yes">/home/<your-username>/Downloads</directories>
```

Code Snippet 1: Directories to be checked by FIM

Save the file and restart the agent:

```
sudo systemctl restart wazuh-agent
```

Code Snippet 2: Restarting agent command

4.2 Enable VirusTotal Integration & Register Script on the Manager

Add the following section inside the <ossec_config> block in the Wazuh Manager:

```
<integration>
  <name>virustotal</name>
  <api_key>ecc4*****89</api_key>
  <group>syscheck</group>
  <alert_format>json</alert_format>
</integration>
<command>
  <name>remove-threat</name>
  <executable>remove-threat.sh</executable>
  <timeout_allowed>no</timeout_allowed>
</command>
<active-response>
  <disabled>no</disabled>
  <command>remove-threat</command>
  <location>local</location>
  <rules_id>87105</rules_id>
</active-response>
```

Code Snippet 3: VirusTotal API Key Block of Code and registering the Script

Full documentation reference: [VirusTotal integration](#)

4.3 Create the Malware Removal Script on the Agent

On the agent machine, create the Active Response script:

```
sudo subl /var/ossec/active-response/bin/remove-threat.sh
```

Code Snippet 4: Creating the malware removal file

The script for this malware removal file is:

```
#!/bin/bash

# Go to Wazuh base directory
LOCAL_DIR="$(dirname "$0")"
cd "$LOCAL_DIR/.." || exit 1
BASE_DIR="$(pwd)"

LOG_FILE="$${BASE_DIR}/../logs/active-responses.log"

# Read JSON input from Wazuh (one line)
read INPUT_JSON

# Extract file path and command from JSON
FILENAME=$(echo "$INPUT_JSON" | jq -r '.parameters.alert.data.virustotal.source.file')
COMMAND=$(echo "$INPUT_JSON" | jq -r '.command')

# Only act for "add" (when AR is triggered)
if [ "$COMMAND" = "add" ]; then
    # Ask wazuh-execd if we can continue (standard AR handshake)
    printf '{"version":1,"origin":{"name":"remove-threat","module":"active-response"},"command":"check_keys","parameters":{"keys":[]}}\n'

    read RESPONSE
    DECISION=$(echo "$RESPONSE" | jq -r '.command')

    if [ "$DECISION" != "continue" ]; then
        echo "$(date '+%Y/%m/%d %H:%M:%S') remove-threat.sh: Aborted by excd: $INPUT_JSON" >> "$LOG_FILE"
        exit 0
    fi
fi
```

Code Snippet 5: Script for Malware removal

Make the script executable:

```
sudo chmod 750 /var/ossec/active-response/bin/remove-threat.sh  
sudo chown root:wazuh /var/ossec/active-response/bin/remove-threat.sh
```

Code Snippet 6: Script to be executable

Restart the manager:

```
sudo systemctl restart wazuh-manager
```

Code Snippet 7: Restarting the Wazuh Manager

Restart the Wazuh Agent:

```
sudo systemctl restart wazuh-agent
```

Code Snippet 8: Restarting the Wazuh agent

Advanced users can build more response scripts using: [Custom active response scripts](#)

5. Testing the Automatic Removal (Using EICAR)

For testing, I used a file from **EICAR**, which is a trusted organization providing test samples for antivirus software. The EICAR test file is safe and used worldwide for security testing. This file can be downloaded using the following command from EICAR in the Wazuh agent:

```
curl -Lo ~/Downloads/suspicious-file.com https://secure.eicar.org/eicar.com
```

Code Snippet 9: EICAR source file

This file is designed to mimic malware behavior, so VirusTotal will mark it as suspicious immediately.

6. Active Response Results in Wazuh Dashboard (Linux Agent)

Once the integration is enabled, Wazuh automatically sends file hashes to VirusTotal whenever FIM detects a new or modified file. The results can be viewed directly in the Wazuh Dashboard.

To check the VirusTotal alerts:

Open the **Wazuh Dashboard** -> Go to **Threat Hunting**.

In the search bar, type **virustotal**

There will be all events generated by the VirusTotal integration, including the file path, detection results, and the number of antivirus engines that flagged the file.

Below is an example screenshot from the test that has been performed using the EICAR test file:

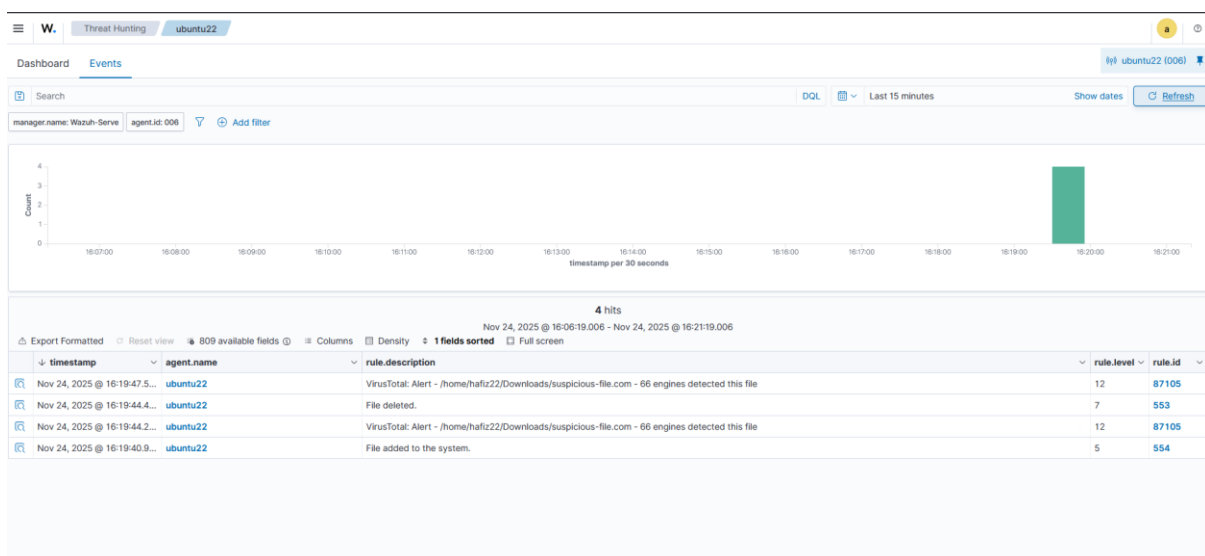


Figure 2: Wazuh Dashboard response in Threat Hunting

Learn how to analyze threats using the Threat Hunting module: [Threat hunting](#)

7. Windows Support (Windows Agent)

The workflow described in this tutorial is not limited to Linux. Wazuh **File Integrity Monitoring (FIM)**, **VirusTotal integration**, and **Active Response** can also be used to protect **Windows endpoints**.

By monitoring directories such as:

C:\Users*\Downloads

C:\Users*\Documents

C:\Users*\Desktop

Code Snippet 10: Directories to be monitored

Wazuh detects new or modified files and sends their hashes to VirusTotal. If VirusTotal identifies the file as malicious, Active Response can automatically trigger a custom script (compiled as remove-threat.exe) that deletes the threat before it has a chance to execute.

For a complete Windows walkthrough including FIM configuration, VirusTotal checks, and generating an executable removal script, please follow the official guide below:

[Ransomware protection on Windows with Wazuh](#)

8. Conclusion

With this setup:

- **FIM** detects new or modified files
- **VirusTotal** checks the file's reputation
- **Active Response** automatically deletes or quarantines malware
- All events appear immediately in the **Wazuh Dashboard**

This workflow transforms Wazuh from passive monitoring into **real-time automated protection**, reducing response time and improving overall system security.